

Data flow Simple (one way) Half-duplex (bidirection, one at a time) Full-duplex (two way at a time)
Network Criteria : performance (time), reliability (failure)
Point to point direct, multipoint bus
Physical topology, mesh, star (home lan, hub), bus (many stops until end), Ring topology (bus connected end)
LAN, WAN (countries, state, international)
Internetwork : connected router
Network 4 problems, connectivity, addressing, forwarding, reliability

Layer	The Specialist	The "Job"
Application	The "Writer"	You type an email. This layer just cares about the text/message.
Transport	The "Supervisor"	Chops the message into pieces and gives them numbers (1 of 5, 2 of 5).
Network	The "Navigator"	Puts the pieces in envelopes and writes the IP Addresses on them.
Data Link	The "Driver"	Moves the envelope from one specific point to the next (e.g., Laptop to Router).
Physical	The "Road"	Converts everything into electrical pulses or light flashes on a wire.

OSI : Application : session, presentation, application
Period = 1/frequency | Phase = position of wave form relative to 0
Analog = continuous, Digital = discrete
wavelength = speed/f = a distance simple signal can travel in one period
Bandwidth : Difference between lowest and highest frequencies in the signal. **Digital signal** : discrete, **Analog** : continuous **Periodic** means there is a pattern **Bit rate (digital)** : describe non-periodic signal. **Bit length** : 1/bit rate. **Digital signal** : composite analog signal with $0 < \text{freq} < \text{inf}$ **Baseband** : send digital signal with no change **Broadband** : change to analog then send.
Signal impairment : **Attenuation** loss of power, medium resistance $10 \log_{10} (\text{power received} / \text{power sent}) = \text{negative dB}$ | **Distortion** signal change form or shape, in a **composite signal** (signal made of different frequencies), **Noise** | **SNR (signal to noise ratio)** = $\text{avg sig power} / \text{avg noise power}$ | **Data rate limit** How fast we can send data in bits per second, depends on bandwidth available, level of signal used, quality of the channel (level of noise) **Nyquist bit rate, theoretical bit rate, NOISELESS** = $\text{Bit rate} = 2 * \text{bandwidth} * \log_2 (\text{signal level})$
Shannon capacity NOISY channel data rate limit

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution
First, we use the Shannon formula to find the upper limit.

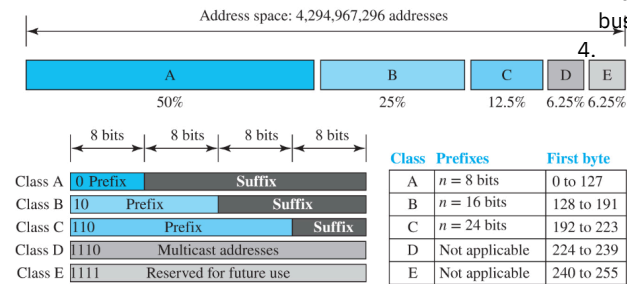
C = B log₂ (1 + SNR) = 10⁶ log₂ (1 + 63) = 10⁶ log₂ 64 = 6 Mbps

The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps, for example. Then we use the Nyquist formula to find the number of signal levels.

4 Mbps = 2 × 1 MHz × log₂ L → log₂ L = 2 → L = 4

SNRdB : $10 * \log_{10} (\text{SNR})$
Bandwidth can be used to measure network performance
Throughput : measurement of how fast data can be sent through the network
Latency (delay) : propagation (travel), transmission (push to wire), queueing (buffer), processing delay (unpack)
Bandwidth delay product: network performance metric
Cross sec = bandwidth * length (delay) **Jitter** delays for each packet | **DIGI to DIGI** Line coding (needed) 0 1 to electrical signal of two levels, block coding (not nee) enlarging bit block, scrambling (not nee) | **ANA to DIGI** digital smtm has less noise : Pulse Code Modulation turn it into discrete form, Delta modulation whether signal up or down, and change it into digital format 0 for down 1 for up | **DIGI → ANA** lowpass channel allows signal that starts with 0, bandpass doesn't, Amplitude shift keying, freq, phase, Phase and freq stays the same, if 1 means max amp, if 0 means low to none amp, Phase (point in cycle where wave starts)

ANA → ANA : ANA info to ANA signal. Medium is bandpass, sometimes ANA can start at 0 too | **AM** Amplitude modulation, carrier amplitude now match modulating sig change, freq and phase modulation
Multiplexing : combine several low bandwidth ch. to yield one large high bandwidth channel **Frequency division multiplexing** : wasted bandwidth, too large, so we divide to share **Time-division multiplexing** concurrent signal transmission, take turns **Transmission media** electromag, light, electrical **3Khz to 1Ghz Radio, 1 to 300 Ghz Micro, 300 Ghz to 400 THz, infared**
Collision simultaneous transmission on shared medium, corrupted frame, collision domain hub = 1 switch 1 port, modern internet isolate every link and run full-duplex, collision with port is possible, hub is a broadcaster, one shared medium
CSMA/CD Listen, send if available, detect collision if collide send jam signal, every device back off for random period of time and join again
Switch MAC learning + unicast forwarding, **flooding** unknown MAC address, broadcast to everyone
Escape subnet, packet with MAC address of a router
Store and forward, waits until entire frame arrives, error detection using CRC
Cut-through, doesn't wait, no CRC, but can cause garbage to be sent **Buffering Flow control** : PAUSE signal when buffer is full **Segregation** port pairs can talk concurrently **Occasional collision** : Half-duplex, duplex-mismatch **Switch fabric**, port isolation, **VLAN** groups ports together, **ALOHA** everyone talks, if fail, try again. **CSMA/CA Wifi**, Listen, if clear then wait for some more to be sure, if not, pause count down, *RTS and CTS will be here for large files and busy traffic*, send data, wait for ACK to report completion, if none, assume that it fails to send. **NAV** stay silent to avoid collision. **Hidden node** : nodes that are not connected but hear someone else's collision, **Exposed node** : hear someone else frame and stay silent unnecessarily, not every tranmission you hear will actually interfere, **BSS coloring / spatial reuse** for concurrent transmission. **MU-MIMO** multiple antennas to signal in different direction **OFDMA** empty truck to full truck that stops many times, one data silent, to full data, pauses and go. **Ethernet frame**, Destination MAC, Source MAC, Ethertype : IPv4 IPv6 what is inside a payload, MTU (max trans unit) **ARP Broadcast to ask, reply (unicast)** with MAC address. **Spoofing and ARP poison** : hacker disguises as router. **Modern day** : switch CSMA/CA wifi **Packetizing** : carry packet from source to destination, encapsulate packet and decapsulate, **Flow control** amount of data source cna send without overwhelming destination, **Congestion control** : too many datagrams concentrated in one area of internet **Data gram : Virtual Circuit Approach** lay out path virtually and leads packets to that path. **Delay** trans = packet length / transmission rate, time taken to put bits in packet Propagate travel delay = distance / propagation file procession **Packet loss** : Buffer full, packet is dropped. **IP** address of connection **Address space** : total no. of addresses used by protocol, IPv4 binary, dotted-decimal, hexadecimal notation.



Access the text alternative for slide images.

- The Two Main "Logic" Families
Before a protocol is made, we need an algorithm.
 - Distance-Vector (DV): Think of this as "Gossip." Routers only talk to their immediate neighbors and share their best guesses on how far away things are. It uses the Bellman-Ford equation to constantly update these guesses.
 - Link-State (LS): Think of this as "Map-Making." Every router floods the network with information about its own direct connections. Eventually, every router has an identical map of the whole network and runs Dijkstra's Algorithm to find the shortest path.
 - Intra-Domain vs. Inter-Domain
The Internet is too big for one protocol, so we divide it into Autonomous Systems (AS) (like an ISP or a University).
 - Intra-domain (Interior): Routing inside your own building/company.
 - RIP: Uses DV logic. It's simple but slow. It counts "Hops" (max 15).
 - OSPF: Uses LS logic. It's very fast and smart. It uses costs like "Bandwidth" or "Delay" to pick paths.
 - Inter-domain (Exterior): Routing between different companies (e.g., True to AIS).
 - BGP: Uses Path-Vector logic. It doesn't just look at distance; it looks at the "Path" (a list of AS numbers) to prevent loops and follows business policies (e.g., "Don't send my data through a rival company's network").
 - Layer 2 vs. Layer 3 Trees
 - STP (Spanning Tree Protocol): This is for Switches (Layer 2). Its only job is to turn off extra cables to prevent "Loops" where data circles forever and crashes the network.
 - Multicast Trees: Instead of sending 1,000 separate copies of a video stream to 1,000 people (Unicast), the router sends one copy and splits it only when the path divides. This saves a massive amount of bandwidth.
- Summary Cheat-Sheet Version:
- DV (RIP): Simple gossip; uses Bellman-Ford; limited to 15 hops.
 - LS (OSPF): Complex map; uses Dijkstra; picks path by speed/cost.
 - Path-Vector (BGP): The "Internet Protocol"; picks path by AS-list and business rules.
 - STP: For Layer 2 switches; kills loops by blocking redundant paths.
 - Multicast: Efficiency for streaming; data only goes to those who "join" the group.